COMPREHENSIVE GRADUATE REFERENCE NOTES ON THE REAL LINE ($\mathbb{R}$)

(Set-Theoretic, Topological, Sequential, Series, Functional, and Measure-Theoretic Facts)

PART I: CARDINALITY & SET-THEORETIC STRUCTURE OF $\mathbb{R}$

1. Cardinality of $\mathbb{R}$

- **Fact**: $|\mathbb{R}| = \mathfrak{c} = 2^{\aleph_0}$.
- **Detailed Explanation**: This is proved via the binary (or decimal) expansion of real numbers in the interval $[0,1]$. Each real number corresponds to an infinite sequence of 0s and 1s (with a technical identification for dyadic rationals, which are countable and don't affect cardinality). The set of all such sequences has cardinality $|\{0,1\}^{\mathbb{N}}| = 2^{\aleph_0}$. Hence, the continuum $\mathfrak{c}$ is precisely the cardinality of the power set of the natural numbers.

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2. The Continuum Hypothesis (CH)

- **Fact**: The statement "there is no set A with $\aleph_0 < |A| < \mathfrak{c}$" is independent of ZFC.
- **Detailed Explanation**: Gödel (1940) proved that CH cannot be disproved from ZFC (it holds in the constructible universe L), and Cohen (1963) proved that CH cannot be proved from ZFC (using forcing). At the PG level, you must appreciate that $\mathfrak{c}$ is just *one* possible value of the continuum; it can be $\aleph_1$, $\aleph_2$, or even larger, depending on the set-theoretic model.

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3. Cardinality of all non-degenerate intervals

- **Fact**: Every interval (a,b), $[\mathrm{a},\mathrm{b}]$, $(a,b]$, etc., has cardinality $\mathfrak{c}$.
- **Detailed Explanation**: The standard bijection is $f(x) = \tan\left(\pi\left(x - \frac{1}{2}\right)\right)$ which maps $(0,1)$ homeomorphically onto $\mathbb{R}$. Since linear maps send any open interval to $(0,1)$, all intervals are equinumerous to $\mathbb{R}$. Length is a measure-theoretic concept, not a set-theoretic one.

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4. Countable vs. uncountable dense subsets

- **Fact**: $\mathbb{Q}$ is countable ($\aleph_0$) and dense, while $\mathbb{R} \setminus \mathbb{Q}$ is uncountable ($\mathfrak{c}$) and dense.

- **Detailed Explanation**: This highlights a crucial topological lesson: **density does not imply largeness in cardinality, nor does it imply "bigness" in the Baire category sense**. Both are pervasive throughout the real line, yet one is negligibly small (countable) and the other is massive (uncountable).

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5. The Cantor set $\mathcal{C}$

- **Fact**: The Cantor set has cardinality $\mathfrak{c}$, yet it contains no interval and has empty interior.
- **Detailed Explanation**: Via ternary expansion, $\mathcal{C}$ is homeomorphic to $\{0,2\}^{\mathbb{N}}$, which has cardinality $2^{\aleph_0} = \mathfrak{c}$. This is the classic counterexample showing that **uncountable does not mean "large" topologically**. It is totally disconnected and perfect.

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6. The Cantor–Bendixson theorem

- **Fact**: Every closed set $F \subset \mathbb{R}$ can be uniquely decomposed as $F = P \cup C$, where P is perfect and C is countable.
- **Detailed Explanation**: The proof iteratively removes isolated points using the *derived set* F' (set of limit points). After countably many (or transfinitely many up to ω_1) iterations, the process stabilizes to a perfect set. The removed points are at most countable. This is a foundational structure theorem in descriptive set theory.

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7. Cardinalities of Borel vs. Lebesgue measurable sets

- **Fact**: The Borel σ-algebra has cardinality $\mathfrak{c}$, while the Lebesgue σ-algebra has cardinality $2^{\mathfrak{c}}$.
- **Detailed Explanation**: The Borel sets are generated by a countable family of intervals through countable unions/intersections/complements—there are only $\mathfrak{c}$ many such sets. However, the Lebesgue completion adds all subsets of measure-zero sets. There are $2^{\mathfrak{c}}$ subsets of the Cantor set (which has measure zero), and all of them are Lebesgue measurable. Hence, **most Lebesgue measurable sets are not Borel**.

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8. Number of continuous functions

- **Fact**: The set $C(\mathbb{R}, \mathbb{R})$ has cardinality $\mathfrak{c}$.

- **Detailed Explanation**: A continuous function is completely determined by its values on the countable dense subset $\mathbb{Q}$. Hence, $|C(\mathbb{R},\mathbb{R})| \leq |\mathbb{R}^{\mathbb{Q}}| = \mathfrak{c}^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0} = \mathfrak{c}$. The lower bound is obvious (constant functions). In stark contrast, the set of all arbitrary functions $\mathbb{R} \to \mathbb{R}$ has cardinality $2^{\mathfrak{c}}$.

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9. Cantor–Schröder–Bernstein theorem

- **Fact**: This theorem is the primary tool for proving equicardinality without constructing explicit bijections.
- **Detailed Explanation**: It states: if there is an injection from A to B and an injection from B to A, then there is a bijection. Using this, one easily proves $|\mathbb{R}^n| = \mathfrak{c}$ for any finite n, and crucially, $|\mathbb{R}^{\mathbb{N}}|$ (the set of all real sequences) also equals $\mathfrak{c}$, because $\mathfrak{c}^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph_0}$.

PART II: TOPOLOGICAL STRUCTURE – OPEN, CLOSED, DENSE, NOWHERE DENSE, AND THE BOREL HIERARCHY (Facts 10–18)

10. Structure of open sets

- **Fact**: Every non-empty open subset of $\mathbb{R}$ is a countable disjoint union of open intervals.
- **Detailed Explanation**: For each x in an open set U, take the maximal interval $(a_x, b_x) \subset U$ containing x. These maximal intervals are pairwise disjoint and there can only be countably many of them because each contains a distinct rational number. This theorem **fails** in general topological spaces (e.g., $\mathbb{R}^2$).

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11. Structure of closed sets

- **Fact**: A closed set is the complement of a countable union of disjoint open intervals.
- **Detailed Explanation**: Equivalent to the statement: every closed set is a G_δ set (countable intersection of open sets) because its complement is F_σ. For example, $[0, 1]$ is closed, and $\mathbb{R} \setminus [0,1] = (-\infty, 0) \cup (1, \infty)$.

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12. The Borel hierarchy

- **Fact**: Open sets are Σ_1^0, closed are Π_1^0; F_σ (countable unions of closed) are Σ_2^0, G_δ (countable intersections of open) are Π_2^0.

- **Detailed Explanation**: $\mathbb{Q}$ is F_σ (a countable union of singletons, which are closed). However, $\mathbb{Q}$ is **not** G_δ in $\mathbb{R}$. Why? If $\mathbb{Q}$ were G_δ, it would be a dense G_δ, hence comeager by Baire's theorem. But $\mathbb{Q}$ is meager (countable union of nowhere dense sets). Contradiction. This is a standard PG exercise.

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13. Separability and Second-countability

- **Fact**: $\mathbb{R}$ is separable ($\mathbb{Q}$ is dense) and second-countable (has a countable base: intervals with rational endpoints).
- **Detailed Explanation**: These properties imply that $\mathbb{R}$ is **hereditarily Lindelöf** (every open cover of any subspace has a countable subcover) and hereditarily separable. This makes $\mathbb{R}$ a very "manageable" space for analysis.

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14. Nowhere dense sets and Baire category

- **Fact**: A set is nowhere dense if its closure has empty interior.
- **Detailed Explanation**: The Cantor set is nowhere dense, as is $\mathbb{Z}$. $\mathbb{Q}$ is not nowhere dense because $\bar{\mathbb{Q}} = \mathbb{R}$ has non-empty interior. The **Baire Category Theorem** states that $\mathbb{R}$ is not a countable union of nowhere dense sets. This is the foundational theorem for *generic* properties.

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15. Meager vs. Comeager sets

- **Fact**: $\mathbb{Q}$ is meager (first category), while $\mathbb{R} \setminus \mathbb{Q}$ is comeager (residual).
- **Detailed Explanation**: Although $\mathbb{Q}$ is dense, it is topologically "small" in the Baire sense. The irrationals, despite being dense and comeager, form a **Baire space** in their own right (with a suitable complete metric). This illustrates the vast gap between category and density.

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16. The irrationals as a Baire space

- **Fact**: $\mathbb{R} \setminus \mathbb{Q}$ is completely metrizable (admits a complete metric) and hence a Baire space.
- **Detailed Explanation**: The standard Euclidean metric on $\mathbb{R} \setminus \mathbb{Q}$ is not complete. However, one can construct a complete metric equivalent to the usual topology by leveraging a homeomorphism to $\mathbb{N}^{\mathbb{N}}$ (Baire space) via continued fractions. This is a classic PG-level construction.

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17. Boundary, interior, and closure identities

- **Fact**: For any $A \subset \mathbb{R}$, $\partial A = \bar{A} \cap \bar{A^c}$, and $\bar{A} = \mathrm{Int}\,(A) \cup \partial A$.
- **Detailed Explanation**: A is open iff $A = \mathrm{Int}\,(A)$; closed iff $A = \bar{A}$. These identities are used constantly in proofs. For $A = \mathbb{Q}$, $\partial A = \mathbb{R}$ and $\mathrm{Int}\,(A) = \emptyset$.

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18. Advanced Cantor set properties

- **Fact**: $\mathcal{C}$ is homeomorphic to the Cantor space $\{0,1\}^{\mathbb{N}}$. Moreover, every compact metric space is a continuous image of $\mathcal{C}$.
- **Detailed Explanation**: The first part is proved via ternary expansions. The second is the **Cantor–Hausdorff theorem** (also called the Cantor space universal property). This makes the Cantor set the "universal compact metric space" for continuous surjections, a cornerstone in general topology.

PART III: SEQUENCES – CONVERGENCE, LIMSUP/LIMINF, AND SUBSEQUENCES (Facts 19–24)

19. Cauchy criterion (Completeness)

- **Fact**: A sequence (x_n) in $\mathbb{R}$ converges iff it is Cauchy.
- **Detailed Explanation**: This is the analytical definition of completeness. $\mathbb{Q}$ fails this (e.g., decimal approximations to $\sqrt{2}$ are Cauchy in $\mathbb{Q}$ but do not converge in $\mathbb{Q}$). This property is equivalent to the Least Upper Bound property.

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20. Bolzano–Weierstrass theorem (sequential form)

- **Fact**: Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.
- **Detailed Explanation**: Proof uses the nested interval principle: bisect the interval containing the sequence infinitely many times. This is equivalent to compactness of closed bounded intervals.

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21. Limsup and Liminf

- **Fact**: $\limsup\ x_n = \inf_{n\geq 1} \sup_{k\geq n} x_k$ and $\liminf\ x_n = \sup_{n\geq 1} \inf_{k\geq n} x_k$.

- **Detailed Explanation**: They always exist in $\mathbb{R} \cup \{\pm\infty\}$. A sequence converges to L iff $\limsup\ x_n = \liminf\ x_n = L$. They are the supremum and infimum of the set of all subsequential limits (cluster points).

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22. Cluster points (subsequential limits)

- **Fact**: The set of all subsequential limits of a bounded sequence is a non-empty closed set.
- **Detailed Explanation**: Its maximum is the limsup, and its minimum is the liminf. For example, the sequence $(-1)^n(1 + 1/n)$ has cluster points -1 and 1. The set $\{-1,1\}$ is closed.

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23. Monotone sequence theorem

- **Fact**: Every bounded monotone sequence converges to its supremum (if increasing) or infimum (if decreasing).
- **Detailed Explanation**: This is a direct consequence of the Least Upper Bound property. It is the most elementary convergence test and is frequently used to define $e = \lim\ (1 + 1/n)^n$.

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24. Monotone subsequence lemma

- **Fact**: Every sequence of real numbers has a monotone subsequence.
- **Detailed Explanation**: Proof uses the concept of "peaks" (indices where the term is greater than or equal to all subsequent terms). If there are infinitely many peaks, they form a decreasing subsequence. If there are finitely many, after the last peak, one can construct an increasing subsequence. This lemma is fundamental for proving Bolzano–Weierstrass.

PART IV: INFINITE SERIES – CONVERGENCE TESTS AND REARRANGEMENTS (Facts 25–34)

25. Necessary condition for convergence

- **Fact**: If $\sum a_n$ converges, then $a_n \to 0$.
- **Detailed Explanation**: This is necessary but not sufficient. The harmonic series $\sum 1/n$ is the standard counterexample: terms go to 0, but the series diverges.

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26. Absolute convergence

- **Fact**: Absolute convergence ($\sum |a_n| < \infty$) implies convergence ($\sum a_n$ converges).
- **Detailed Explanation**: Proof uses the Cauchy criterion for series. The converse is false: the alternating harmonic series $\sum(-1)^{n+1}/n$ converges conditionally (not absolutely).

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27. Riemann rearrangement theorem

- **Fact**: For a conditionally convergent series $\sum a_n$, for any $L \in [-\infty, \infty]$, there exists a permutation π such that $\sum a_{\pi(n)} = L$ (or diverges to $\pm\infty$).
- **Detailed Explanation**: This is one of the most counterintuitive PG-level results. It exploits the fact that the positive and negative terms each diverge, allowing you to "add enough positives" to overshoot L, then "add enough negatives" to undershoot, etc. In contrast, absolutely convergent series are permutation-invariant.

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28. Cauchy product (Mertens' theorem)

- **Fact**: If $\sum a_n$ converges absolutely to A and $\sum b_n$ converges to B, their Cauchy product converges to AB.
- **Detailed Explanation**: If both converge conditionally, the product may fail to converge. For example, the Cauchy product of $\sum(-1)^n/\sqrt{n+1}$ with itself diverges. Absolute convergence of at least one series is essential.

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29. Dirichlet's test

- **Fact**: If (a_n) has bounded partial sums and (b_n) decreases monotonically to 0, then $\sum a_n b_n$ converges.
- **Detailed Explanation**: This generalizes the alternating series test (take $a_n = (-1)^n$). It is a workhorse for studying trigonometric series like $\sum \sin(nx)/n$.

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30. Abel's test

- **Fact**: If $\sum a_n$ converges and (b_n) is bounded and monotone, then $\sum a_n b_n$ converges.

- **Detailed Explanation**: This is the dual of Dirichlet's test. It is used to prove Abel's limit theorem (Fact 34) and in studying power series at the boundary of the radius of convergence.

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31. Cauchy condensation test

- **Fact**: For a non-increasing positive sequence a_n, $\sum a_n$ converges iff $\sum 2^k a_{2^k}$ converges.
- **Detailed Explanation**: This test proves convergence/divergence of $\sum 1/(n(\log\ n)^p)$. For $p > 1$, it converges; for $p \leq 1$, it diverges. It is highly efficient for log-based series.

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32. Raabe's and Gauss's tests

- **Fact**: Raabe's test: if $n(1 - a_{n+1}/a_n) \to L$, then $\sum a_n$ converges if $L > 1$, diverges if $L < 1$.
- **Detailed Explanation**: When the ratio test gives $L = 1$ (borderline), Raabe's test refines it. Gauss's test goes further by comparing to a harmonic series, handling cases where Raabe's limit is exactly 1 (e.g., $\sum 1/(n(\log\ n)^p)$).

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33. Radius of convergence (Hadamard's formula)

- **Fact**: For a power series $\sum c_n (x-a)^n$, the radius of convergence is $R = 1/\limsup\ |c_n|^{1/n}$.
- **Detailed Explanation**: This is the definitive formula. Inside the interval $|x - a| < R$, the series converges absolutely; outside, it diverges. At the boundary, convergence depends on the specific coefficients (e.g., $\sum x^n/n$ converges at $x = -1$ but not at $x = 1$).

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34. Abel's limit theorem

- **Fact**: If $\sum a_n$ converges to S, then $\lim_{x \to 1^-} \sum a_n x^n = S$.
- **Detailed Explanation**: This bridges discrete series and continuous functions. It is used to evaluate sums like $\sum (-1)^{n+1}/n = \ln\ 2$ by taking the limit $x \to 1^-$ of $\sum (-1)^{n+1} x^n/n = \ln\ (1 + x)$.

PART V: FUNCTIONS – CONTINUITY, UNIFORMITY, DISCONTINUITIES, BAIRE CLASSES (Facts 35–43)

35. Sequential vs. ε-δ continuity

- **Fact**: A function is continuous at x_0 iff for every sequence $x_n \to x_0$, we have $f(x_n) \to f(x_0)$.
- **Detailed Explanation**: This equivalence holds for metric spaces (and first-countable spaces). It is often easier to use in proofs (e.g., proving that addition of continuous functions is continuous).

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36. Heine–Cantor theorem (Uniform continuity on compact sets)

- **Fact**: Every continuous function on a compact interval $[a, b]$ is uniformly continuous.
- **Detailed Explanation**: This fails on non-compact sets. Example: $f(x) = x^2$ on $\mathbb{R}$ is continuous but not uniformly continuous, because ε-δ cannot work globally for large x (the needed δshrinks as | x | grows).

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37. Discontinuities of monotone functions

- **Fact**: A monotone function has at most countably many discontinuities, all of which are jump discontinuities.
- **Detailed Explanation**: At each discontinuity, there is a jump of size > 0. Since the total variation over an interval is bounded, the sum of all jumps is finite, so there can only be countably many jumps. The Cantor function (devil's staircase) is continuous, monotone, and has uncountably many flat regions—yet its derivative is 0 a.e.

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38. Classification of discontinuities

- **Fact**: Discontinuities are classified as removable (limit exists but not equal), jump (one-sided limits exist but differ), or essential (at least one one-sided limit does not exist).
- **Detailed Explanation**: A major PG-level theorem: the set of discontinuities of any real-valued function is an F_σ set. Conversely, every F_σ set can be realized as the discontinuity set of some function. This is a deep connection between topology and function theory.

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39. Darboux property of derivatives

- **Fact**: Derivatives possess the Intermediate Value Property, even if they are not continuous.
- **Detailed Explanation**: This means a derivative cannot have a simple jump discontinuity. For example, the function $g(x) = 0$ for $x \neq 0$ and $g(0) = 1$ is not a derivative, as it violates IVT. The standard proof uses the Flett or Darboux theorem via the mean value theorem.

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40. Baire class one functions

- **Fact**: A Baire class one function is the pointwise limit of continuous functions. Such functions have points of continuity on every closed interval (Baire's theorem).
- **Detailed Explanation**: The Dirichlet function $1_{\mathbb{Q}}$ is not Baire class one (its discontinuities are all of $\mathbb{R}$, which is not meager). It is Baire class two (pointwise limit of Baire class one functions). The Baire hierarchy extends through all countable ordinals.

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41. Bounded variation, absolute continuity, and the Cantor function

- **Fact**: Functions of bounded variation decompose as the difference of two increasing functions (Jordan decomposition). Absolute continuity implies differentiability a.e. (Lebesgue theorem).
- **Detailed Explanation**: The Cantor function is continuous, of bounded variation, and maps the measure-zero Cantor set onto $[0, 1]$. It is **not** absolutely continuous because absolutely continuous functions send measure-zero sets to measure-zero sets. The Cantor function violates this.

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42. Lipschitz and Hölder continuity

- **Fact**: Lipschitz functions ($| f(x) - f(y) | \leq C | x - y |$) are absolutely continuous and differentiable a.e. (Rademacher's theorem in 1D).
- **Detailed Explanation**: Hölder with exponent $\alpha < 1$ is weaker than Lipschitz. Example: $f(x) = \sqrt{x}$ is Hölder with $\alpha = 1/2$ on $[0, 1]$ but not Lipschitz. The space of Hölder functions forms a Banach space and is central in PDE theory.

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43. Nowhere differentiable continuous functions

- **Fact**: There exist continuous functions that are differentiable at no point (e.g., Weierstrass function).
- **Detailed Explanation**: The Weierstrass function $f(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$ with $0 < a < 1$ and $ab > 1 + 3\pi/2$ is continuous everywhere but differentiable nowhere. Moreover, in the space $C([0,1])$ with the sup-norm, such functions are **generic** (they form a dense G_δ set) by Baire's theorem.

PART VI: THE GRAND THEOREMS OF REAL ANALYSIS (Facts 44–53)

44. Least Upper Bound Property (Completeness)

- **Fact**: Every non-empty subset of $\mathbb{R}$ bounded above has a supremum in $\mathbb{R}$.
- **Detailed Explanation**: This is the foundational axiom that distinguishes $\mathbb{R}$ from $\mathbb{Q}$. It underpins all of calculus: without it, the IVT, EVT, and convergence of Cauchy sequences all fail.

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45. Nested Intervals Theorem (Cantor)

- **Fact**: For a decreasing sequence of closed, bounded, non-empty intervals $[a_n, b_n]$ with lengths $\to 0$, the intersection is a singleton.
- **Detailed Explanation**: This theorem is equivalent to completeness. It is used to prove Bolzano–Weierstrass and the uncountability of $\mathbb{R}$.

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46. Heine–Borel Theorem

- **Fact**: $K \subset \mathbb{R}$ is compact (open-cover definition) iff it is closed and bounded.
- **Detailed Explanation**: In arbitrary metric spaces, "closed and bounded" is necessary but not sufficient for compactness (e.g., the closed unit ball in infinite-dimensional Banach spaces is not compact). Heine–Borel is special to $\mathbb{R}^n$ and finite-dimensional normed spaces.

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47. Bolzano–Weierstrass (Set form)

- **Fact**: Every bounded infinite subset of $\mathbb{R}$ has a limit point.

- **Detailed Explanation**: This is equivalent to the sequential compactness of closed bounded intervals. It forms the basis of the Weierstrass Approximation Theorem via Bernstein polynomials.

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48. Intermediate Value Theorem (IVT)

- **Fact**: Continuous functions map intervals to intervals.
- **Detailed Explanation**: If f is continuous on $[a, b]$ and k lies between $f(a)$ and $f(b)$, then $\exists c \in (a, b)$ with $f(c) = k$. This is the topological result that the continuous image of a connected set is connected, and connected subsets of $\mathbb{R}$ are exactly intervals.

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49. Extreme Value Theorem (EVT)

- **Fact**: A continuous function on a compact interval attains its maximum and minimum.
- **Detailed Explanation**: The image of a compact set under a continuous function is compact (closed and bounded in $\mathbb{R}$). Hence, the image contains its supremum and infimum. This fails on open intervals (e.g., $f(x) = x$ on $(0, 1)$).

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50. Dini's Theorem

- **Fact**: If (f_n) is a pointwise increasing sequence of continuous functions on a compact set converging pointwise to a continuous f, then convergence is uniform.
- **Detailed Explanation**: The monotonicity is crucial. Without it, pointwise convergence on compact sets need not be uniform (e.g., $f_n(x) = x^n$ on $[0, 1]$ converges pointwise to a discontinuous limit, but adjust it to remove discontinuity—still, monotonicity is key).

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51. Arzelà–Ascoli Theorem

- **Fact**: A subset of $C([a, b])$ is compact iff it is uniformly bounded and equicontinuous.
- **Detailed Explanation**: This is the foundational theorem for existence of solutions to ODEs (Peano's existence theorem). Equicontinuity ensures that a

subsequence converges uniformly. It generalizes the Bolzano–Weierstrass theorem to function spaces.

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52. Baire Category Theorem – Consequences

- **Fact**: $\mathbb{R}$ is of second category. Consequently, the set of points of continuity of a Baire-one function is a dense G_δ.
- **Detailed Explanation**: Baire's theorem is the engine behind many "generic" results. For instance, a generic continuous function is nowhere differentiable. Also, a generic $f \in C([0,1])$ is not of bounded variation.

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53. Lindelöf property

- **Fact**: Every open cover of any subset of $\mathbb{R}$ has a countable subcover.
- **Detailed Explanation**: This follows from second-countability. It makes $\mathbb{R}$ a **paracompact** space. Para compactness is essential for constructing partitions of unity in differential geometry, proving that smooth manifolds admit Riemannian metrics.

PART VII: MEASURE-THEORETIC AND DESCRIPTIVE SET-THEORETIC INSIGHTS

54. Lebesgue outer measure

- **Fact**: The Lebesgue measure of an interval is its length. $\mathbb{Q}$ has measure zero, the Cantor set has measure zero, and $\mathbb{R}$ has infinite measure.
- **Detailed Explanation**: The Cantor set is the classic example of an uncountable set with measure zero. It shows that cardinality ($\mathfrak{c}$) and measure are completely unrelated. A countable union of measure-zero sets has measure zero (subadditivity), which proves $\mathbb{Q}$ has measure zero.

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55. Non-measurable sets (Vitali sets)

- **Fact**: Assuming the Axiom of Choice, there exist subsets of $\mathbb{R}$ that are not Lebesgue measurable.
- **Detailed Explanation**: Vitali's construction uses the equivalence relation $x \sim y$ iff $x - y \in \mathbb{Q}$. Choosing one representative from each coset of $\mathbb{R}/\mathbb{Q}$ yields a set V that, when translated by rationals, partitions $[0, 1]$. If V were measurable, it

would lead to a contradiction ($m([0,1])$ being either 0 or infinite). This result justifies the need for measurable spaces and sigma-algebras.

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56. Fat Cantor set (Smith–Volterra–Cantor)

- **Fact**: A nowhere dense, closed, uncountable subset of $[0,1]$ can have **positive Lebesgue measure**.
- **Detailed Explanation**: To construct it, remove middle intervals of lengths that shrink sufficiently slowly (e.g., removing $1/4, 1/16, 1/64, \ldots$). The total length removed is $1/2 < 1$, so the leftover has measure $1/2$. This is a brutal reminder that **nowhere dense does not imply measure zero**—a crucial distinction for PG students.

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57. Borel vs. Analytic vs. Projective sets

- **Fact**: Borel sets form the smallest σ-algebra containing open sets. Analytic sets (continuous images of Borel sets) are universally measurable but need not be Borel.
- **Detailed Explanation**: The projection of a Borel set in $\mathbb{R}^2$ onto $\mathbb{R}$ is analytic. Under the Axiom of Choice + V=L, there exist analytic sets that are not Borel. This enters the realm of **descriptive set theory**. The projective hierarchy extends this with complements and projections, leading to independence results (e.g., the existence of sets without the Baire property).

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58. Cantor–Bendixson rank

- **Fact**: For any closed set F, the iterated derived set (removing isolated points) stabilizes at some countable ordinal $\alpha < \omega_1$.
- **Detailed Explanation**: This rank provides a fine classification of closed subsets. For example, the Cantor set has rank 1 (it is its own derived set). Countable ordinals arise as ranks, showing the deep connection between topology and transfinite induction. This theorem is a beautiful introduction to the use of ordinals in analysis.